

COMPUTER SCIENCES

CS 368-1 (2011 Summer) — Day 6 Homework

Due Thursday, July 21, at the start of class.

Description

Starting at a given directory, find all of the directories contained inside recursively. Then, for each directory found, display the directory name and information about the files found inside. To be successful, you will use many built-in functions and modules.

Details

The script will take one directory path as input — either ask the user for the directory or simply hard-code it into your script. Then, starting at that directory, search for and store all directories contained inside (including the given one) *recursively*.

Then, go through the directories that you found and display a report for each one:

- List all of the regular files in the directory (not directories!) — how do you tell a regular file from other kinds of files?
- Sort files by file size (in bytes) from largest to smallest — How do you get file sizes? How do you sort by size (which is numeric)?
- For each file, display the file size, the (formatted) last modification time, and the name of the file (without the directory prefix) — see the example output below for a better picture of what to do. How do you format a date/time? How can you control output formatting in general?

You may assume that there are only regular files and directories.

Example Output

```
Directory </p/course/cs368-cat/public/html/2010>
  9641  2010-07-01 16:25:22  resources.html
  7068  2010-07-20 13:59:20  index.html
  1541  2009-07-08 09:52:37  cs-logo.gif
   707  2010-07-20 16:16:16  homework-08-ideas.pl
   658  2009-07-10 15:49:32  368-1.css

Directory </p/course/cs368-cat/public/html/2010/01-introduction>
  115281  2010-07-12 10:18:22  cs368-1-2010-summer-01-0712M.pdf
   2369  2010-07-12 09:51:49  homework-01-assignment.html

Directory </p/course/cs368-cat/public/html/2010/02-syntax>
  131814  2010-07-13 09:56:27  cs368-1-2010-summer-02-0713T.pdf
   2974  2010-07-13 10:02:07  homework-02-assignment.html
   642  2010-07-13 16:40:02  in-class-code.html
```

```

Directory </p/course/cs368-cat/public/html/2010/03-collections>
  114186   2010-07-15 09:44:21   cs368-1-2010-summer-03-0715R.pdf
    4144   2010-07-15 09:53:12   homework-03-assignment.html

Directory </p/course/cs368-cat/public/html/2010/04-basic-io>
  1454670   2009-07-13 13:46:40   homework-04.txt
  132377   2010-07-16 10:12:51   cs368-1-2010-summer-04-0716F.pdf
    3389   2010-07-16 10:34:29   homework-04-assignment.html

Directory </p/course/cs368-cat/public/html/2010/05-subroutines>
  113456   2010-07-19 10:08:07   cs368-1-2010-summer-05-0719M.pdf
    3213   2010-07-19 10:30:29   homework-05-assignment.html
    1133   2010-07-19 14:39:02   in-class-code.html

Directory </p/course/cs368-cat/public/html/2010/06-data-structures>
  237215   2009-07-20 23:57:39   example_access_log
  124530   2010-07-20 09:49:22   cs368-1-2010-summer-06-0720T.pdf
    8092   2010-07-20 10:14:27   homework-06-assignment.html

Directory </p/course/cs368-cat/public/html/2010/08-standard-library>
  2484     2010-07-20 16:15:22   homework-08-assignment.html

```

Hints

The following hints are a bit ... vague. I am doing this to encourage you [to look up](#) some functions and modules online. You need to learn to do this anyway, so consider it good practice.

- Look at the `stat()` function, especially for the last modification time
- There are at least two different ways to find the size of a file — either is fine
- All of the built-in functions and modules you need were in today's slides
- I suggest organizing your script in two parts: collecting the list of directories, then reporting on each one

Extra Challenges

- [easy] Sort the directories in string order
- [hard] Write a short **but clear** script! My solution is 31 lines (with normal formatting, including blank lines), but it could be shorter&hellip

Reminders

Do the work yourself, consulting reasonable reference materials as needed; any reference material that gives you a complete or nearly complete solution to this problem or a similar one is not OK to use. Asking the instructors for help is OK, asking other students for help is not.

Hand In

A **printout** of your code on a **single sheet of paper**. Be sure to put your own name in the initial comment block of the code. Identifying your work is important, or you may not receive appropriate credit.